

$$\begin{aligned}
&\equiv [\neg p \vee (p \wedge \neg q)] \vee (p \vee q) \quad (1st \text{ Commutative Law}) \\
&\equiv [(\neg p \vee p) \wedge (\neg p \vee \neg q)] \vee (p \vee q) \quad (1st \text{ ~~Associative~~ Distributive Law}) \\
&\equiv [(p \vee \neg p) \wedge (\neg p \vee \neg q)] \vee (p \vee q) \quad (1st \text{ Commutative Law}) \\
&\equiv [\neg p \wedge (\neg p \vee \neg q)] \vee (p \vee q) \quad (1st \text{ Negation Law}) \\
&\equiv [(\neg p \vee \neg q) \wedge \neg p] \vee (p \vee q) \quad (2nd \text{ Commutative Law}) \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q) \quad (1st \text{ Identity Law}) \\
&\equiv \neg p \vee (\neg q \vee (p \vee q)) \quad (1st \text{ Associative Law}) \\
&\equiv \neg p \vee ((p \vee q) \vee \neg q) \quad (1st \text{ Commutative Law}) \\
&\equiv \neg p \vee (p \vee (q \vee \neg q)) \quad (1st \text{ Associative Law}) \\
&\equiv \neg p \vee (p \vee T) \quad (1st \text{ Negation Law}) \\
&\equiv \neg p \vee T \quad (1st \text{ Domination Law}) \\
&\equiv T \quad (1st \text{ Domination Law})
\end{aligned}$$

$$\begin{aligned}
g) [p \wedge (p \rightarrow q)] \rightarrow q &\equiv [p \wedge (p \rightarrow q)] \vee q \quad (\text{Conditional-disjunction equivalence}) \\
&\equiv [\neg p \vee \neg(p \rightarrow q)] \vee q \quad (1st \text{ De Morgan's Law}) \\
&\equiv [\neg p \vee \neg(\neg p \vee q)] \vee q \quad (\text{Conditional-disjunction equivalence}) \\
&\equiv [\neg p \vee \neg(\neg p) \wedge \neg q] \vee q \quad (2nd \text{ De Morgan's Law}) \\
&\equiv [\neg p \vee (p \wedge \neg q)] \vee q \quad (\text{Double negation law}) \\
&\equiv [\neg p \vee (p \wedge \neg q)] \vee q \quad (1st \text{ Distributive Law}) \\
&\equiv [(\neg p \vee p) \wedge (\neg p \vee \neg q)] \vee q \quad (1st \text{ commutative law}) \\
&\equiv [(p \vee \neg p) \wedge (\neg p \vee \neg q)] \vee q \quad (1st \text{ Negation law}) \\
&\equiv [T \wedge (\neg p \vee \neg q)] \vee q \quad (2nd \text{ commutative Law}) \\
&\equiv [(\neg p \vee \neg q) \wedge T] \vee q \quad (1st \text{ Identity Law}) \\
&\equiv (\neg p \vee \neg q) \vee q \quad (1st \text{ Associative Law}) \\
&\equiv \neg p \vee (\neg q \vee q) \quad (1st \text{ Commutative Law}) \\
&\equiv \neg p \vee (q \vee \neg q) \quad (1st \text{ Negation Law}) \\
&\equiv \neg p \vee T \quad (1st \text{ Domination Law})
\end{aligned}$$